Sustainability Economics: Complete Lecture Notes

John Janmaat
Economics, Philosophy and Political Science, The University of British Columbia

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Course Overview & Syllabus Summary

What is the connection between economics and sustainability? When people talk about economics, often they are talking about something to do with money. How people acquire it and what they do with it. When people talk about sustainability, often they are talking about the environment, how it is changing, and whether or not those changes are good. Where do these two things come together? That intersection often seems to be about the way that people are acquiring and using money is changing the environment in ways that are not good.

Part I

Foundations & Basic Principles

Chapter 1

Introduction & Key Concepts

1.1 Overview & Objectives

The term *sustainability* implies continuity and balance. If something is sustainable, it can continue doing what it is doing. Unless it is doing nothing, what it is doing is interacting with other entities. This implies that, except for those things that are doing nothing, we should think about sustainability with a systems perspective.

A system is a collection of entities (or elements) and the connections between them. To understand a system, we need to understand the entities and their connections. Our goal in this course is to develop an understanding of the relationship between the economy and the environment by looking at the elements that make up the economy and their relationship with elements that make up the environment. To do so, we will proceed by building a model of a *complex adaptive social ecological system*.

The system we inhabit, and that we will try to understand, is complex. This means that the system as a whole exhibits properties that cannot be explained by looking at any element itself. It is adaptive in that it changes in response to internal and external pressures.

ey reading for this section: Chapter 1 of the required textbook.

Systems

- A system is a collection of elements and their connections.
- A social ecological system contains elements in both the social (human) and ecological (nature) domains, and has connections crossing between the domains.
- A system is complex when there are 'emergent' properties that cannot be predicted looking at elements in isolation.
- A system is adaptive if it continuously evolves or changes in response to internal and external forces.

Systems



1.1.1 Detailed Explanation of Concept 1

While the slide highlights the main points, it is important to note that the theoretical model relies on two primary assumptions...

Mathematical Representation

$$f(x) = \alpha + \beta x + \varepsilon$$

Chapter 2

Hermit Economics

2.1 Learning Objectives

- Explain the meaning of opportunity cost.
- Demonstrate an understanding of the production function and the production possibilities frontier.
- Explain the role of diminishing marginal product.

2.2 The Hermit

A hermit is someone who has chosen to live apart from society. In the most extreme case, they have no contact with any other person and are completely self sufficient. From a system thinking perspective, the hermit would be one element of the system, with the other elements being part of the ecological side of the hermit-nature social ecological system.

Hermit System

- Hermit,
- Garden,
- Forest.

Systems

2.2.1 Detailed Explanation of Concept 1

While the slide highlights the main points, it is important to note that the theoretical model relies on two primary assumptions...

Mathematical Representation

$$f(x) = \alpha + \beta x + \varepsilon$$